THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF MATHEMATICS AND STATISTICS

November 2017

MATH3311/MATH5335

Mathematical Computing for Finance
Computational Methods for Finance

(1) TIME ALLOWED – 2 HOURS

(2) TOTAL NUMBER OF QUESTIONS – 4

(3) ANSWER **QUESTIONS 1 AND 2.**
ANSWER **EITHER** QUESTION 3 **OR** QUESTION 4.
(ONLY ONE OF QUESTIONS 3 AND 4 CAN COUNT.)

(4) THE QUESTIONS ARE OF EQUAL VALUE

(5) THIS PAPER MAY BE RETAINED BY THE CANDIDATE

(6) **ONLY** CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [20 marks]

i) Let $A \in \mathbb{R}^{m\times n}$ with $m \geq n$.

a) Write down the singular value decomposition and state the properties of the matrices.

b) Assume that the singular values of the matrix A are given by $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$. Assume that $\sigma_1 = 10^3, \ldots, \sigma_{10} = 100, \sigma_{11} = 10^{-5}, \ldots, \sigma_n = 10^{-12}$. How could you obtain an approximation of the matrix A using the largest 10 singular values?

c) Storing the matrix A requires $8mn$ bytes in double precision. How much storage is required to store your approximation to A in double precision?

ii) Let $Q \in \mathbb{R}^{n\times n}$.

a) What property must Q satisfy to be an orthogonal matrix?

b) Show that the condition number of Q in 2-norm is 1.

iii) Let C be a real symmetric n by n matrix, with ordered eigenvalues

$$\lambda_1 = -2.0 \cdot 10^{-4} \leq \lambda_2 = 4.8 \cdot 10^{-4} \leq \cdots \leq \lambda_n = 2 \cdot 10^5.$$

a) Is the matrix C positive definite? Give a reason for your answer.

b) The MATLAB command det(C) yields 0. Is this correct? Provide an explanation for your answer.

c) Find the condition number of the matrix C in an appropriate norm.

d) Assume we want to solve the linear system $C\mathbf{x} = \mathbf{b}$ in double precision in MATLAB. The elements of C are known exactly, and the computed solution $\mathbf{x}$ must have at least 4 significant figures. How accurate must $\mathbf{b}$ be to achieve this?

iv) Consider the matrix

$$D = B + \mathbf{u}\mathbf{v}^\top,$$

where

$$B = \begin{pmatrix} 1 & 0 & 0 & \ldots & 0 \\ 1 & 1 & 0 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 1 & \ldots & \ldots & 1 & 1 \end{pmatrix}$$

and where $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are column vectors. Assume that n is large (say $n = 10^6$).

a) Why is explicitly forming D not a good idea?

b) Let $\mathbf{z} \in \mathbb{R}^n$ be given. Provide an algorithm which computes $D\mathbf{z}$ using the smallest possible number of additions and multiplications. How many additions and multiplications does your algorithm require to compute $D\mathbf{z}$?

1) i) a) 2 $A = U\begin{bmatrix}\Sigma \\ 0\end{bmatrix} V^T$, $U \in \mathbb{R}^{m\times m}$ orthogonal, $\Sigma \in \mathbb{R}^{n\times n}$ diagonal

$V \in \mathbb{R}^{n\times n}$ orthogonal.

b) 2 $A \approx \underbrace{U(:,1:10)}_{m\times 10} * \underbrace{\Sigma(1:10,1:10)}_{10\times 10} * \underbrace{V(:,1:10)'}_{n\times 10}$

c) 2

To store U we need $m \times 10 \times 8$ bytes
Σ we need 8×10 bytes
V we need $n \times 10 \times 8$ bytes

} need $80(m+n)+80$ bytes

ii) a) 1 $Q \cdot Q^T = I$, identity matrix

b) 2 $Q^{-1} = Q^T$. Let $x \in \mathbb{R}^n$ with $x \cdot x = 1$. Then

$Qx \cdot Qx = x^T Q^T Q x = x^T I x = x^T x = 1$. Thus

$$\|Q\|_2 = \max_{x\neq 0} \frac{\|Qx\|_2}{\|x\|_2} = \max_{\substack{x \neq 0 \\ \|x\|_2 = 1}} \|Qx\|_2 = 1.$$

Further $\|Q^T\|_2 = 1$ (use the same argument to show that)

Thus $K_2 = \|Q\|_2 \, \|Q^{-1}\|_2 = \|Q\|_2 \, \|Q^T\|_2 = 1$.

iii) 1 a) No, the first eigenvalue is negative

1 b) No, $\det(C) = \lambda_1 \cdot \lambda_2 \cdots \lambda_n < 0$ since all eigenvalues are nonzero.

2 c) $K_2 = \dfrac{2\cdot 10^5}{2\cdot 10^{-4}} = 10^9$. symmetric?

2 d) $\text{relerr}(x) \lesssim K_2(A)\,(\underbrace{\text{relerr } C}_{2.2\cdot 10^{-16}} + \text{relerr}(b))$; $\text{relerr}(x) < \frac{1}{2}\cdot 10^{-4}$

$$\text{relerr}(b) \approx \frac{\text{relerr}(x)}{K_2(A)} - \text{relerr}(C) \approx \tfrac{1}{2}\cdot 10^{-13} - 2.2\cdot 10^{-6} < \tfrac{1}{2}\cdot 10^{-13}.$$

$\Rightarrow$ 13 significant figures in b are required.

iv) a) The matrix B has a simple structure and is sparse.
(2) Forming D destroys both properties. and takes up lot of memory!

b)
(3) $D\underline{z} = B\underline{z} + \underline{u}\,\underline{v}^T\underline{z}$. Let $\underline{z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}$.

Computing Bz:

Let $B\underline{z} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$. Then
$$\begin{aligned} a_1 &= z_1 \\ a_2 &= z_1 + z_2 \\ &\vdots \\ a_n &= z_1 + \dots + z_n \end{aligned}$$

We can compute this in the following way:

$a_1 = z_1$
for $k = 2:n$
  $a_k = z_k + a_{k-1}$; $n-1$ additions, 0 multiplications
end

Compute $\underline{v}^T \cdot \underline{z} =: b$

$\underline{v}^T \cdot \underline{z} = v_1 z_1 + v_2 z_2 + \dots + v_n z_n$; requires $n-1$ additions, n multiplicati

Compute $b \cdot \underline{u} = \begin{pmatrix} b u_1 \\ b u_2 \\ \vdots \\ b u_n \end{pmatrix}$. Requires n multiplications

Now compute final solution:

$$\begin{pmatrix} a_1 + b u_1 \\ \vdots \\ a_n + b u_n \end{pmatrix}$$; requires n additions.

Number of additions: $3n-2$
Number of multiplications: $2n$.

2. [20 marks]

i) A deterministic interest model $r(s)$ at time $s \in [0, T]$ interpolates the $n = 9$ data values in Table 1.

i	1	2	3	4	5	6	7	8	9
t_i (years)	0	0.5	1	1.5	2	2.5	3	3.5	4
$r(t_i)$ (%)	3.9	3.6	3.4	3.8	4.1	4.4	5.1	4.9	4.7

Table 1: Interest rate data

a) Consider interpolating the interest data using a polynomial.

A) How high does the degree of the polynomial have to be to ensure that the data can be interpolated?

B) Is using a polynomial a good idea? Give reasons for your answer.

b) The following MATLAB commands were used to generate Figure 1.

```
t = 0:0.5:4;
r = [3.9 3.6 3.4 3.8 4.1 4.4 5.1 4.9 4.7];
tp = linspace(-0.5, 4.5, 1001);
pp1 = spline(t,[1 r -1],tp);
pp2 = spline(t,r,tp);
plot(t,r,'.',tp,pp1,':r',tp,pp2,'--b');
legend('Data','Spline 1', 'Spline 2');
```

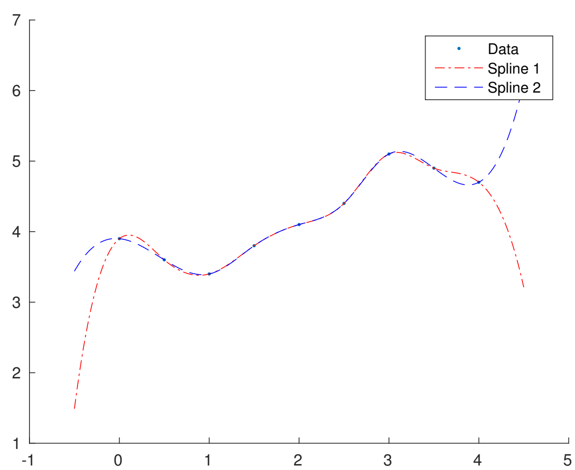

Figure 1: Interest rate data and Spline approximations

A) State the definition of a cubic spline.

B) Explain the properties of Spline 1.

C) Explain the properties of Spline 2.

ii) It is known that for any $0 \le n \neq m < N$ we have

$$\sum_{k=0}^{N-1} \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right)\cos\left(\frac{\pi}{N}\left(k+\frac{1}{2}\right)\left(m+\frac{1}{2}\right)\right) = 0$$

and for $n = m$ we have

$$\sum_{k=0}^{N-1} \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right)\cos\left(\frac{\pi}{N}\left(k+\frac{1}{2}\right)\left(n+\frac{1}{2}\right)\right) = \frac{N}{2}.$$

(You do NOT have to proof these relations.)

In some applications the fast Fourier transform is replaced by the discrete cosine transform. One version of this transform is defined as follows: Let $x_0, x_1, \ldots, x_{N-1} \in \mathbb{R}$ be unknown values and let

$$y_k = \sum_{n=0}^{N-1} x_n \cos\left(\frac{\pi}{N}\left(n+\frac{1}{2}\right)\left(k+\frac{1}{2}\right)\right) \quad \text{for } k = 0, 1, \ldots, N-1. \tag{1}$$

Assume you are given $y_0, y_1, \ldots, y_{N-1}$. Find the inverse discrete cosine transform of (1) which allows you to compute $x_0, x_1, \ldots, x_{N-1}$. Provide a proof for your answer.

iii) Let $f \in C^2([a, b])$ and $x \in (a, b)$. Consider the expression

$$\frac{f(a+h) - f(a)}{h} = f'(a) + O(h). \tag{2}$$

a) Use Taylor series expansions to prove (2).

b) An analyst estimates the first derivative of a function using MATLAB at $a = 2 * 10^6$ using $h = 10^{-12}$ and gets the value 0. Is this correct?

iv) The standard normal cumulative distribution function (cdf) $\Phi : \mathbb{R} \to [0, 1]$ is

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \mathrm{e}^{-y^2/2} \, \mathrm{d}y. \tag{3}$$

Assume that $x \ge 0$. In the following we want to estimate $\Phi(x)$ using a Gauss-Legendre rule.

The MATLAB command

```
[z, w] = gauleg(N);
```

calculates the N Gauss-Legendre nodes z_j, $j = 1, 2, \ldots, N$ and weights w_j, $j = 1, 2, \ldots, N$ for the interval $[-1, 1]$.

Assume we want to estimate the integral (3) with an error at most $\varepsilon > 0$.

a) Find a value of $b < 0$ such that

$$\int_{-\infty}^{b} \mathrm{e}^{-y^2/2} \, \mathrm{d}y < \varepsilon/2.$$

Note that any value $b < 0$ which satisfies the inequality will be sufficient.

b) Explain how the Gauss-Legendre nodes and weights can be used to approximate the integral

$$\int_{b}^{x} \mathrm{e}^{-y^2/2} \, \mathrm{d}y.$$

(Q2) i) a) A) degree 8 1

B) No; oscillating 1

b) A) piecewise cubic poly. 2

twice differentiable everywhere

2 B) derivative $s'(0)=1$; $s'(4)=-1$

1 c) not a knot spline: $s_1'''(0.5) = s_2'''(0.5)$

$s_7'''(3.5) = s_8'''(3.5)$

ii)
4

$$\frac{2}{N}\sum_{k=0}^{N-1} y_k \cos\left(\frac{\pi}{N}(m+\tfrac{1}{2})(k+\tfrac{1}{2})\right)$$

$$=\frac{2}{N}\sum_{k=0}^{N-1}\sum_{n=0}^{N-1} x_n \cos\left(\frac{\pi}{N}(n+\tfrac{1}{2})(k+\tfrac{1}{2})\right)\cos\left(\frac{\pi}{N}(m+\tfrac{1}{2})(k+\tfrac{1}{2})\right)$$

$$=\frac{2}{N}\sum_{n=0}^{N-1} x_n \sum_{k=0}^{N-1}\cos\left(\frac{\pi}{N}(n+\tfrac{1}{2})(k+\tfrac{1}{2})\right)\cos\left(\frac{\pi}{N}(m+\tfrac{1}{2})(k+\tfrac{1}{2})\right)$$

$$=\frac{2}{N}\sum_{n=0}^{N-1} x_n \begin{Bmatrix} \frac{N}{2} & \text{if } n=m \\ 0 & \text{if } n\neq m \end{Bmatrix} = x_m.$$

iii) a) We have

2

$$f(a+h) = f(a) + h f'(a) + O(h^2) \quad \text{for } f\in C^2$$

Thus $$\frac{f(a+h)-f(a)}{h} = f'(a) + O(h)$$

b) Let $a = 2\cdot 10^6$ and $h = 10^{-12}$. Thus we have $a+h = 2\cdot 10^6 + 10^{-12}$

2 $= 2\cdot 10^6 \left(1 + \underbrace{\tfrac{1}{2} 10^{-18}}_{< eps \approx 2.2\cdot 10^{-16}}\right)$. Since $\frac{1}{2}\cdot 10^{-18} < eps$, $1+\frac{1}{2}\cdot 10^{-18}$ will be stored as 1 in double precision. Hence $f(a) - f(a+h) = f(a) - f(a)$ $= 0$. $\Rightarrow$ The result 0 has not been computed correctly.

Q2) iv) a) Assume $b \leq -1$. Then

(2) $$\int_{-\infty}^{b} e^{-y^2/2}\, dy < \int_{-\infty}^{b} e^{-\frac{|y|}{2}}\, dy = \int_{-\infty}^{b} e^{y/2}\, dy = 2e^{b/2} =$$

$$\iff b = 2\log\frac{\varepsilon}{4}$$

b) Using the transformation

$$z = -1 + 2\frac{y-b}{x-b} \iff y = b + \frac{z+1}{2}(x-b)$$

we obtain

(3) $$\int_{b}^{x} e^{-\frac{y^2}{2}}\, dy = \frac{x-b}{2}\int_{-1}^{1} e^{-\frac{1}{2}\left(b+\frac{z+1}{2}(x-b)\right)^2}\, dz.$$

Estimating the last integral by a Gauss-Legendre rule we get

$$\int_{b}^{x} e^{-\frac{y^2}{2}}\, dy = \frac{x-b}{2}\int_{-1}^{1} e^{-\frac{1}{2}\left(b+\frac{z+1}{2}(x-b)\right)^2}\, dz$$

$$\approx \frac{x-b}{2}\sum_{j=1}^{N} w_j\, e^{-\frac{1}{2}\left(b+\frac{z_j+1}{2}(x-b)\right)^2}$$

ANSWER EITHER QUESTION 3 OR QUESTION 4, NOT BOTH

3. [20 marks]

i) Consider a real-valued random variable Y with a continuous, strictly increasing, cumulative distribution function F, so that

$$\mathbb{P}(Y \le y) = F(y).$$

Show that the random variable $X = F(Y)$ is uniformly distributed in the interval $[0, 1]$.

ii) Transform the expected value

$$\mathbb{E}[h] = \int_{-\infty}^{\infty} h(x)\phi(x)\,\mathrm{d}x,$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$ is the standard normal density, into an integral over $[0, 1]$.

iii) An asset price S_t is assumed to follow a geometric Brownian motion

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma \mathrm{d}W_t, \quad t \in (0, T], \tag{4}$$

where W_t is a Wiener process, μ, σ are positive constants and S_0 is given.

a) What are the key properties of the Wiener process W_t?

b) Equation (4) can be discretized using

$$\frac{\Delta S_n}{S_n} = \mu \Delta t + \sigma \sqrt{\Delta t} Z_n,$$

where $Z_0, Z_1, \ldots, Z_{N-1}$ are independent, standard normal random variables, $t_n = n\Delta t$, $\Delta t = \frac{T}{m}$, and $\Delta S_n = S_{n+1} - S_n$.
Assume that you can generate independent samples $u_1, u_2, \ldots$ which are uniformly distributed in the interval $[0, 1]$. How can you use these samples to simulate the asset prices S_t at times t_n for $n = 0, 1, \ldots, m$?

iv) The Koksma-Hlawka inequality

$$\left| \int_{[0,1]^d} f(\mathbf{x})\,\mathrm{d}\mathbf{x} - \frac{1}{N} \sum_{n=1}^{N} f(\mathbf{x}_n \right| \le D(\mathbf{x}_1, \ldots, \mathbf{x}_N) V(f)$$

relates the quadrature error to the discrepancy $D(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ of the point set $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ and the variation $V(f)$ of the function f.

a) What is a 'low-discrepancy' sequence?

b) Why would you use a low-discrepancy sequence in preference to a Monte Carlo method?

c) Explain the idea of variance reduction.

d) How would you use variance reduction in conjunction with quasi-Monte Carlo?

4

3) i) We have $F(y) = \mathbb{P}(Y \leq y)$. Then for $0 \leq a < b \leq 1$

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a \leq F(Y) \leq b)$$

$$= \mathbb{P}(F(Y) \leq b) - \mathbb{P}(F(Y) \leq a)$$

$$= \mathbb{P}(Y \leq F^{-1}(b)) - \mathbb{P}(Y \leq F^{-1}(a))$$

F is strictly increasi
so F^{-1} exists;

$$= F(F^{-1}(b)) - F(F^{-1}(a))$$

$$= b - a.$$

Hence X is uniformly distributed;

ii) We have

3

$$\mathbb{E}(h) = \int_{-\infty}^{\infty} h(x)\,\phi(x)\,dx. \quad \text{Let } \Phi(y) = \int_{-\infty}^{y} \phi(x)\,dx$$

and Φ^{-1} be the inverse of Φ.

Let

$x = \Phi^{-1}(z) \Leftrightarrow z = \Phi(x)$, then $\frac{dz}{dx} = \phi(x)$. Using this substitution we get

$$E(h) = \int_{0}^{1} h(\Phi^{-1}(z))\,dz.$$

iii) a) A Wiener process on $[0,T]$ is a random variable $W(t)$ satisfying

-) $W(t)$ depends continuously on t for $0 \le t \le T$.
-) $W(0) = 0$ with probability 1
-) $W(t) - W(s) \sim N(0, t-s)$ for $0 \le s < t \le T$.
-) $W(t) - W(s)$ and $W(v) - W(u)$ are independent for $0 \le s < t < u < v \le T$.

4

b) Let Φ^{-1} be the inverse standard normal ~~pd~~ c.d.f. Then let $Z_n = \Phi^{-1}(u_{n+1})$ for $n = 0, 1, \dots, N-1$. Let S_0 be given.

3 For $n = 1 : N$ compute

$$S_n = S_{n-1}\left(1 + \mu \Delta t + \sigma \sqrt{\Delta t}\, Z_n\right).$$

iv) a) A low-discrepancy ~~point set~~ sequence $\{x_1, x_2, \dots\}$ is a ~~point set~~ sequence in $[0,1]^s$

1 such that the discrepancy of the first N points $\{x_1, x_2, \dots, x_N\}$ satisfies

$$D(\{x_1, \dots, x_N\}) \le C_s \frac{(\log N)^s}{N}, \quad \text{for all } N \in \mathbb{N},$$

where C_s does not depend on N.

b) A low-discrepancy sequence yields a faster convergence rate

1 of the integration error if the problem/integrand is in some sense low dimensional. The Monte Carlo method only yields a convergence rate of $N^{-1/2}$ compared to $(\log N)^s/N$ for low discrepancy sequences.

c) Assume we want to estimate an integral

$$\int_0^1 f(x)\,dx.$$

Then use a function such that $\int_0^1 g(x)\,dx$ is known

2 explicitely and such that the variance of $f-g$ is small compared $\cancel{\int_0^1 (f(x)-g(x))\,dx}$ to the variance of f.

Then calculate $\int_0^1 g(x)\,dx$ and estimate

$$\int_0^1 (f(x)-g(x))\,dx.$$

d) Variance reduction can be used in the same way as for MC (as in Part c); the only difference is that

2 one would choose g such that the variation (not variance) $V(f-g)$ is small.

ANSWER EITHER QUESTION 3 OR QUESTION 4, NOT BOTH

4 **3.** The value $V(S,t)$ of an option on an underlying asset with price S at time t satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{\sigma^2}{2}S^2\frac{\partial^2 V}{\partial S^2} = rV \tag{3.1}$$

for $0 < S < \infty$ and $0 < t < T$. We assume that the risk-free interest rate r and the volatility σ are known positive constants.

i) ~~a)~~ Explain what the "initial" conditions are for a **call** option, which gives the right to buy the underlying asset for the strike price X at expiry T.

Answer The "initial" condition for a financial option occurs at expiry T, where the payoff of the option is specified. For a call option with strike price X the value at expiry is

1

$$V(S,T) = \max(S-X, 0) = \begin{cases} S - X & \text{if } S > X; \\ 0 & \text{otherwise.} \end{cases}$$

ii ~~b)~~ Carefully explain what the boundary conditions are for this problem.

Answer The boundary conditions specify the behaviour of the option as the underlying asset S goes to its extreme values, namely at $S = 0$ and as $S \to \infty$. The boundary conditions apply for time $t \in (0,T)$.

For a call option, and a model using geometric Brownian motion, if the asset value becomes 0, then it will be 0 for all subsequent times. Thus, when $S = 0$ a call option is worthless, so

$$V(0,t) = 0, \quad t \in (0,T).$$

2

As S gets larger it will lead to an asset price at expiry above the strike, so

$$V(S,t) \to S - Xe^{-r(T-t)} \quad \text{as } S \to \infty.$$

Other possibilities are to specify

$$\lim_{S\to\infty} \frac{\partial V(S,t)}{\partial S} = 1, \qquad \text{or} \qquad \lim_{S\to\infty} \frac{\partial^2 V(S,t)}{\partial S^2} = 0.$$

iii ~~ii)~~ The change of variables to log-moneyness y and time to expiry τ is

$$y = \log(S/X) \qquad \text{and} \qquad \tau = T - t.$$

This gives $W(y,\tau) = V(S,t)$.

a) Find expressions for

$$\frac{\partial W}{\partial \tau}, \qquad \frac{\partial W}{\partial y}, \qquad \frac{\partial^2 W}{\partial y^2},$$

in terms of the partial derivatives of $V(S,t)$.
Answer The variables are related by

$$y = \log(S/X) \iff S = Xe^y, \qquad \tau = T - t \iff t = T - \tau.$$

Hence, as $W(y,\tau) = V(S,t)$,

3

$$\begin{aligned}
\frac{\partial W}{\partial \tau} &= \frac{\partial V}{\partial t}\frac{\partial t}{\partial \tau} = -\frac{\partial V}{\partial t},\\
\frac{\partial W}{\partial y} &= \frac{\partial V}{\partial S}\frac{\partial S}{\partial y} = S\frac{\partial V}{\partial S},\\
\frac{\partial^2 W}{\partial y^2} &= \frac{\partial}{\partial y}\left(S\frac{\partial V}{\partial S}\right) = \frac{\partial}{\partial S}\left(S\frac{\partial V}{\partial S}\right)\frac{\partial S}{\partial y} = \left(\frac{\partial V}{\partial S} + S\frac{\partial^2 V}{\partial S^2}\right)S.
\end{aligned}$$

b) Show that the Black and Scholes PDE (3.1) is equivalent to

$$-\frac{\partial W}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\frac{\partial^2 W}{\partial y^2} = rW \tag{3.2}$$

in terms of the log-moneyness y and the time to expiry τ.
Answer From the expression relating the second derivatives

$$S^2\frac{\partial^2 V}{\partial S^2} = \frac{\partial^2 W}{\partial y^2} - \frac{\partial W}{\partial y}.$$

Substituting the expressions for the derivatives into the Black-Scholes PDE (3.1) gives

2

$$-\frac{\partial W}{\partial \tau} + r\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\left(\frac{\partial^2 W}{\partial y^2} - \frac{\partial W}{\partial y}\right) = rW.$$

Grouping the first derivatives w.r.t y together gives

$$-\frac{\partial W}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\frac{\partial^2 W}{\partial y^2} = rW.$$

c) What are the initial conditions for the transformed problem?
Answer The initial condition for the transformed problem is at $\tau = 0 \iff t = T$, so

1

$$W(y,0) = \max(Xe^y - X, 0) = \max(X(e^y - 1), 0).$$

d) What are the boundary conditions for the transformed problem?
Answer The boundary conditions occur when $S \to 0^+ \iff y \to -\infty$ and $S \to \infty \iff y \to \infty$, giving, for $\tau \in (0,T)$,

2

$\lim_{y\to-\infty} W(y,\tau) = 0, \qquad W(y,\tau) \to X(e^y -$ [redacted] $e^{-r\tau})$ as $y \to +\infty$.

iv) ~~iii)~~ Let $y_{\max} > 0$ and consider the domain $y \in [-y_{\max}, y_{\max}]$ and $\tau \in [0, T]$. Set up a grid

$$y_j = -y_{\max} + j\Delta y,\ j = 0, 1, 2, \ldots, 2(n+1), \quad \tau_\ell = \ell\Delta\tau,\ \ell = 0, 1, 2, \ldots, m,$$

where

$$\Delta y = \frac{y_{\max}}{n+1} \qquad \text{and} \qquad \Delta\tau = \frac{T}{m}.$$

Let $w_j^\ell \approx W(y_j, \tau_\ell)$ denote the approximate solution.

a) At the time point $\tau_{\ell+1}$ and log-moneyness y_j give

A) central difference approximations of $O(\Delta y^2)$ for the partial derivatives

$$\frac{\partial W}{\partial y} \quad \text{and} \quad \frac{\partial^2 W}{\partial y^2}.$$

Answer For partial derivatives w.r.t y, only the y variable, and hence the index j, changes. Using $y_j + \Delta y = y_{j+1}$, $y_j - \Delta y = y_{j-1}$ and $w_j^\ell \approx W(y_j, \tau_\ell)$,

2

$$\left.\frac{\partial W}{\partial y}\right|_{y_j,\tau_{\ell+1}} = \frac{w_{j+1}^{\ell+1} - w_{j-1}^{\ell+1}}{2\Delta y} + O\left(\Delta y^2\right),$$

$$\left.\frac{\partial^2 W}{\partial y^2}\right|_{y_j,\tau_{\ell+1}} = \frac{w_{j+1}^{\ell+1} - 2w_j^{\ell+1} + w_{j-1}^{\ell+1}}{\Delta y^2} + O\left(\Delta y^2\right).$$

B) the backward difference approximation of $O(\Delta\tau)$ for the partial derivative

$$\frac{\partial W}{\partial \tau}.$$

Answer For partial derivatives w.r.t τ, only the τ variable, and hence the index ℓ, changes. Using $\tau_\ell = \tau_{\ell+1} - \Delta\tau$ and $w_j^\ell \approx W(y_j, \tau_\ell)$,

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$$\left.\frac{\partial W}{\partial \tau}\right|_{y_j,\tau_{\ell+1}} = \frac{w_j^{\ell+1} - w_j^\ell}{-\Delta\tau} + O(\Delta\tau) = \underline{\frac{w_j^{\overset{+1}{\ell}} - w_j^{\ell\cancel{+1}}}{\Delta\tau}} + O(\Delta\tau).$$

b) We want to use an **implicit** finite difference method for computing an approximate solution w_j^ℓ to equation (3.2).

A) Show that the finite difference equation has the form

$$\alpha w_{j-1}^{\ell+1} + \beta w_j^{\ell+1} + \gamma w_{j+1}^{\ell+1} = w_j^\ell, \tag{3.3}$$

and find the coefficients α, β and γ.

Answer Ignoring the $O\left(\Delta y^2\right)$ and $O(\Delta\tau)$ terms in the derivative estimates, and substituting in the transformed Black-Scholes PDE

$$\underline{-\frac{\partial W}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial W}{\partial y} + \frac{\sigma^2}{2}\frac{\partial^2 W}{\partial y^2} = rW}$$

gives

$$-\left(\frac{w_j^\ell - w_j^{\ell+1}}{\Delta\tau}\right)+\left(r-\frac{\sigma^2}{2}\right)\left(\frac{w_{j+1}^{\ell+1}-w_{j-1}^{\ell+1}}{2\Delta y}\right)+\frac{\sigma^2}{2}\left(\frac{w_{j+1}^{\ell+1}-2w_j^{\ell+1}+w_{j-1}^{\ell+1}}{\Delta y^2}\right)=rw_j^{\ell+1}.$$

Multiplying through by $-\Delta\tau$ and collecting terms gives

$$\begin{aligned} w_j^\ell &= w_{j-1}^{\ell+1}\left(-\frac{\Delta\tau}{2\Delta y}\left(r-\frac{\sigma^2}{2}\right)+\frac{\sigma^2\Delta\tau}{2\Delta y^2}\right)+ \\ & \quad w_j^{\ell+1}\left(1-r\Delta\tau-\frac{\sigma^2\Delta\tau}{\Delta y^2}\right)+ \\ & \quad w_{j+1}^{\ell+1}\left(\frac{\Delta\tau}{2\Delta y}\left(r-\frac{\sigma^2}{2}\right)+\frac{\sigma^2\Delta\tau}{2\Delta y^2}\right). \end{aligned}$$

2

Thus $\alpha w_{j-1}^{\ell+1}+\beta w_j^{\ell+1}+\gamma w_{j+1}^{\ell+1}=w_j^\ell$ where

$$\begin{aligned} \alpha &= -\frac{\Delta\tau}{2\Delta y}\left(r-\frac{\sigma^2}{2}\right)+\frac{\sigma^2\Delta\tau}{2\Delta y^2}, \\ \beta &= 1-r\Delta\tau-\frac{\sigma^2\Delta\tau}{\Delta y^2}, \\ \gamma &= \frac{\Delta\tau}{2\Delta y}\left(r-\frac{\sigma^2}{2}\right)+\frac{\sigma^2\Delta\tau}{2\Delta y^2}. \end{aligned}$$

B) Briefly explain why it is not a good idea to use an **explicit** method.

Answer In terms of the time variable τ, we are mowing forwards from time $\tau = 0$, where the initial conditions determine the value of the option, to $\tau = T$. Thus a general time step is from τ_ℓ where option values are known, to $\tau_{\ell+1}$ where the option values are unknown.

1

An explicit method has a explicit formula for calculating the unknown variables $w_j^{\ell+1}$ in terms of known values w_j^ℓ. However an explicit method has a stability condition

$$\frac{\Delta\tau}{\Delta y^2} < \rho$$

which restricts the choice of the discretization parameters $\Delta\tau$ and Δy. For example a small value of Δy requires a very small value of $\Delta\tau$ and hence a large number of time steps. If the stability condition is not satisfied then (rounding) errors get amplified until they completely corrupt the numerical solution.

v) ~~iv)~~ Equation (3.3) has the form

$$A\boldsymbol{w}^{\ell+1} = \boldsymbol{w}^\ell, \quad \text{where } \boldsymbol{w}^{\ell+1} = (w_1^{\ell+1},\ldots,w_{2n+1}^{\ell+1})^T.$$

Figure 3.1 shows the spy plot of the coefficient matrix A for $n = 40$.

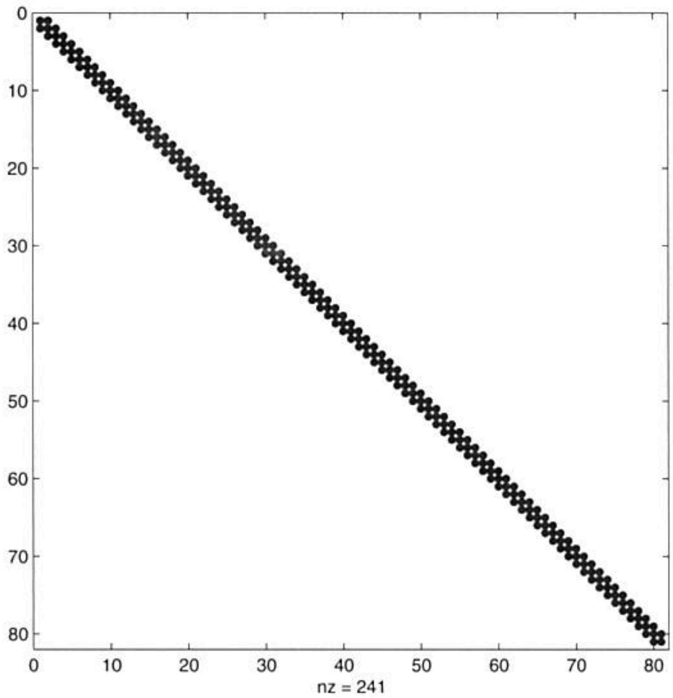

Figure 3.1: Coefficient matrix for one time step of finite difference method

a) A) What are the dimensions of the coefficient matrix A?
Answer The asset discretization is y_j for $j = 0, \ldots, 2(n+1)$, where $y_0 = -y\text{max}$ and $y_{2(n+1)} = y\text{max}$ are the boundaries of the computational domain where the solution is determined by the boundary conditions. Thus the variables are $w_j^{\ell+1}$ for $j = 1, \ldots, 2n+1$. For $n = 40$ the matrix A has dimension $2n+1 = 81$, which agrees with the spy plot in Figure 3.1.

1

B) Calculate the sparsity of A.
Answer The sparsity of A is

$$\frac{\text{nnz}(A)}{n^2} \times 100 = \frac{241}{81^2} \times 100 \approx 3.67\%.$$

1

Here nnz(A), the number of non-zero elements in A, has been read off the spy plot in Figure 3.1.

b) What structure of the matrix A can be exploited to make time-stepping as efficient as possible?
Answer The coefficient matrix A is

- **tridiagonal**, so the upper bandwidth $m_u = 1$ and the lower bandwidth $m_\ell = 1$, and all elements outside the main diagonal, the immediate sub-diagonal and super-diagonal (diagonals $0, -1, 1$) are zero. This means that a linear system with an $\bar{n}$ by $\bar{n}$ coefficient matrix A can be solved in $8\bar{n}$ flops, rather than

1

$O(\bar{n}^3)$ flops when the matrix is treated as a full matrix. This is a very special form of sparsity, and only the $3\bar{n} - 2$ elements on diagonals $-1, 0, 1$ need be stored, rather than $\bar{n}^2$ elements for a full matrix.

- **independent of time** τ, so A can be factored **once**, not on each time step.
- Toeplitz, as the values α, β, γ on the diagonals $-1, 0, 1$ are constant (they do not depend on the index j).

The coefficient matrix A is **not**

- symmetric, as $\alpha \neq \gamma$, so $A_{12} \neq A_{21}$.
- As A is not symmetric, we do not consider whether it is positive definite.